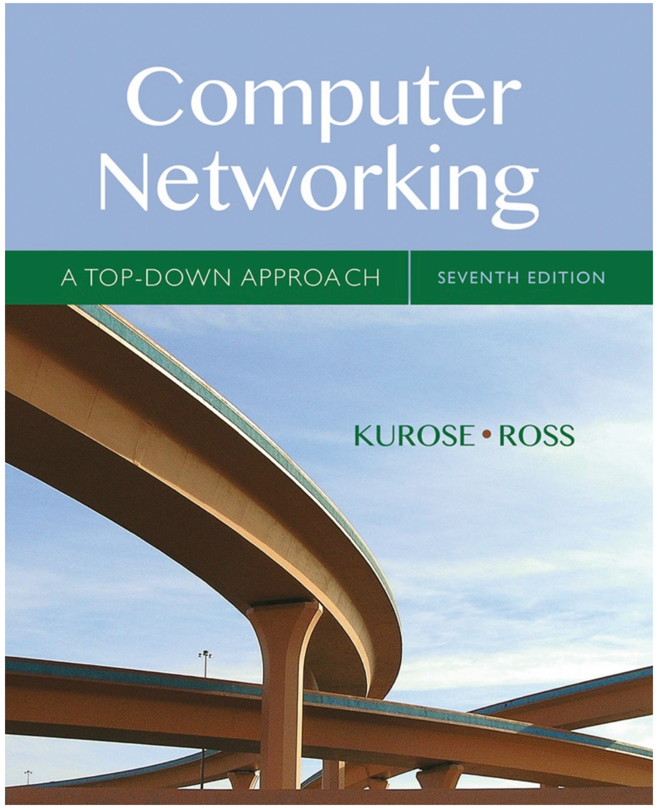
Chapter 6
The Link Layer and LANs



Computer Networking: A Top Down Approach

7th edition
Jim Kurose, Keith Ross
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Thanks and enjoy! JFK/KWR

Link layer, LANs: outline

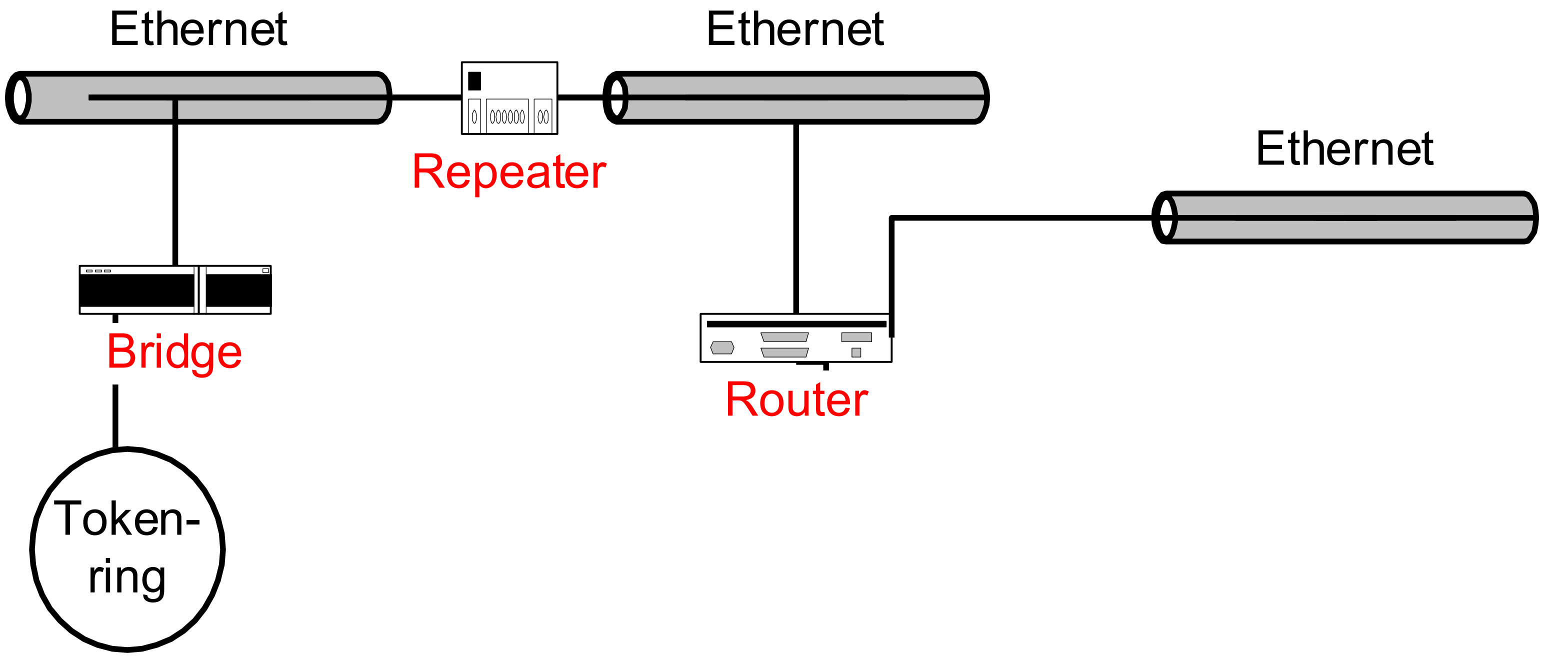
6.4 LAN Switches

- Interconnection devices – Repeaters, Bridges[1], Routers
- Bridges/LAN switches vs. Routers
- Bridges
- Learning Bridges
- Transparent Bridges

[1]bridge: old name for switch

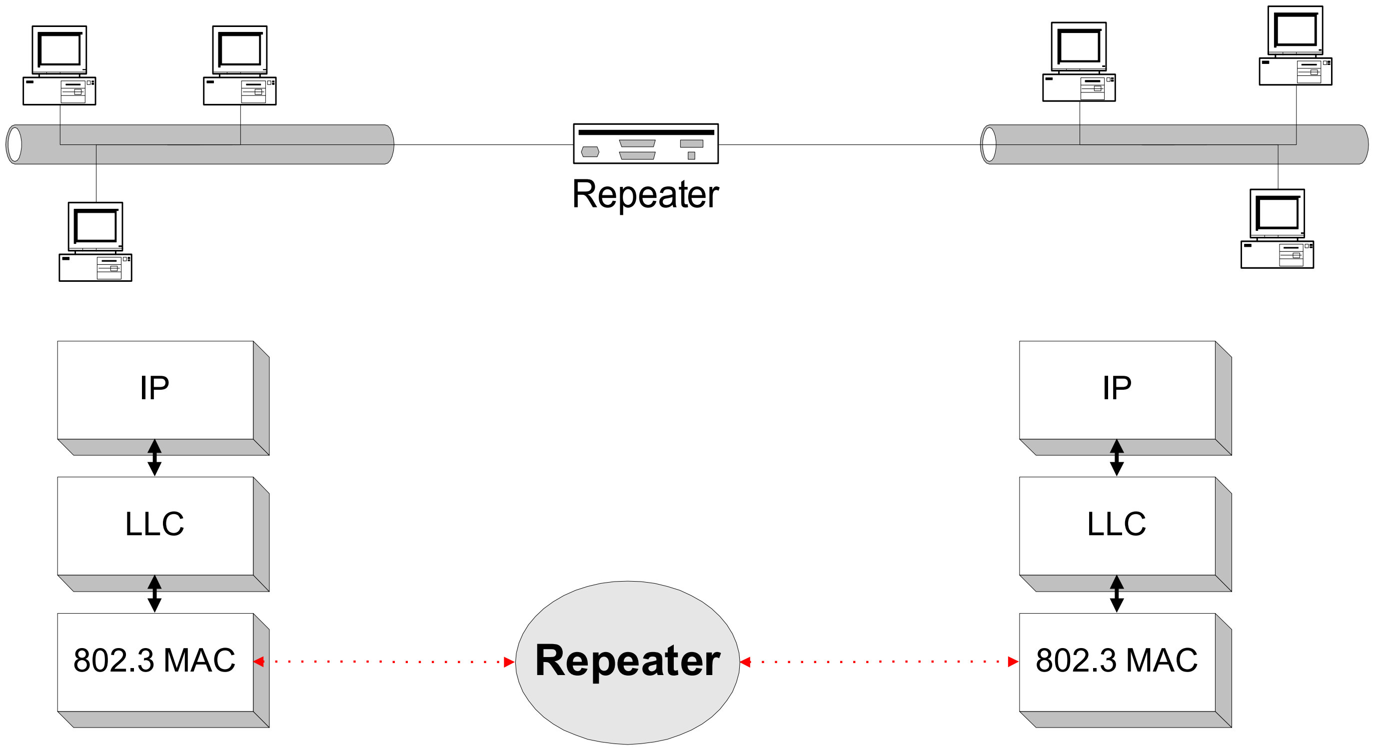
Interconnection Devices

- there are many different devices for interconnecting networks.



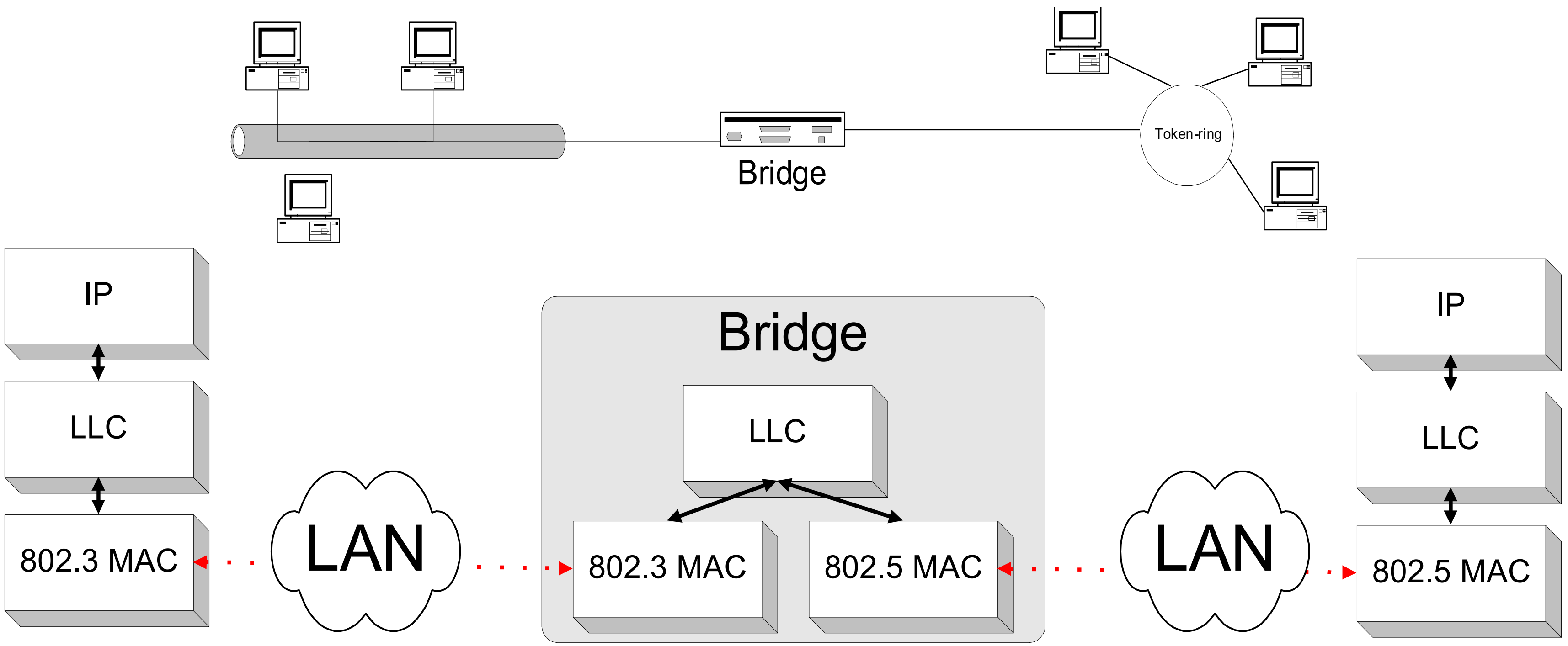
Repeaters

- used to interconnect multiple Ethernet segments
- merely extends the baseband cable
- amplifies all signals including collisions/errors



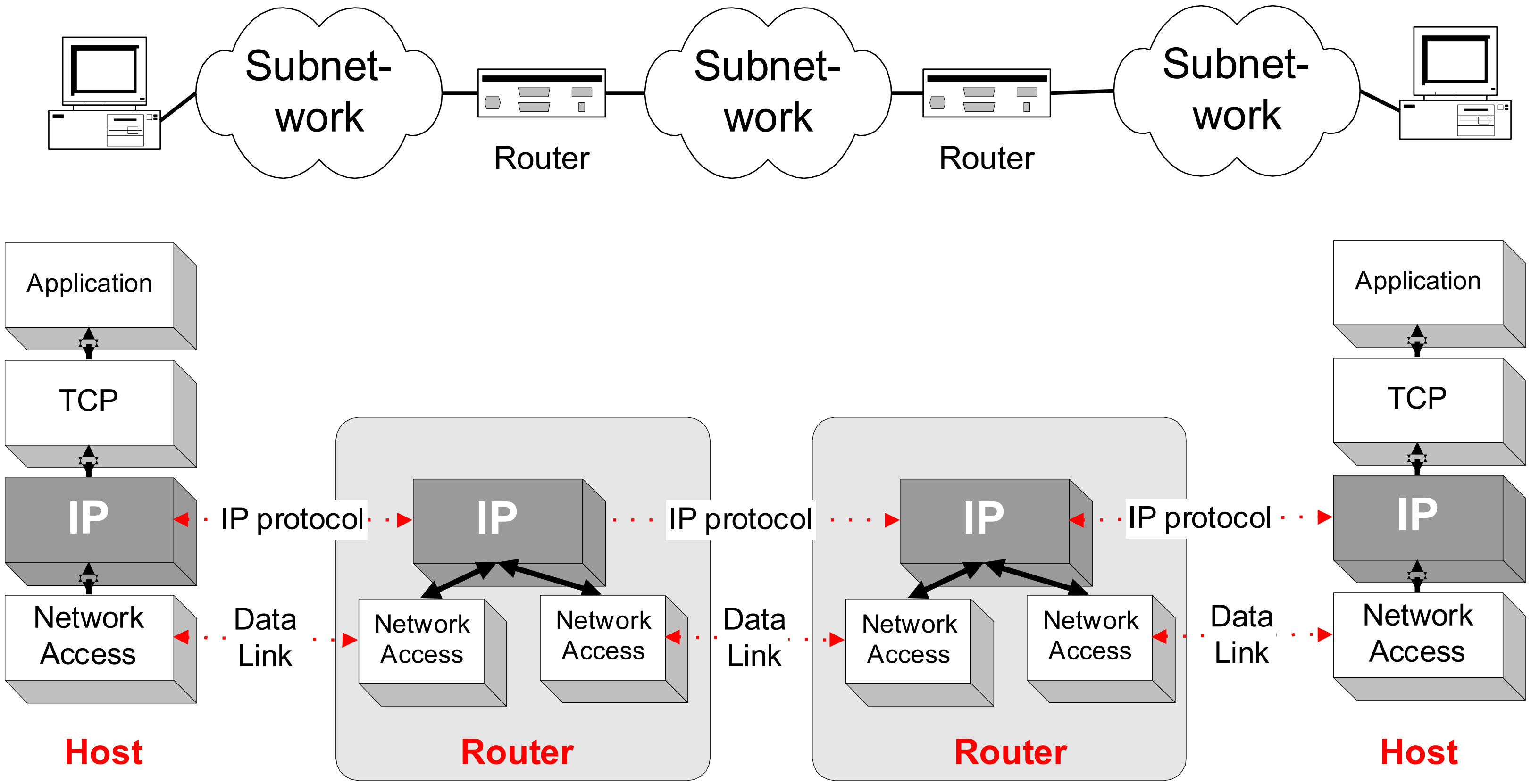
Bridges/LAN switches

- interconnect multiple LANs, possibly different types
- bridges operate at the Data Link Layer (Layer 2) and only forward (switch) link layer frames
- forwarding is done based on MAC addresses and hop-by-hop



Routers

- routers operate at the Network Layer (Layer 3)
- interconnect different subnetworks
- use a forwarding table and IP addresses to route packets hop-by-hop

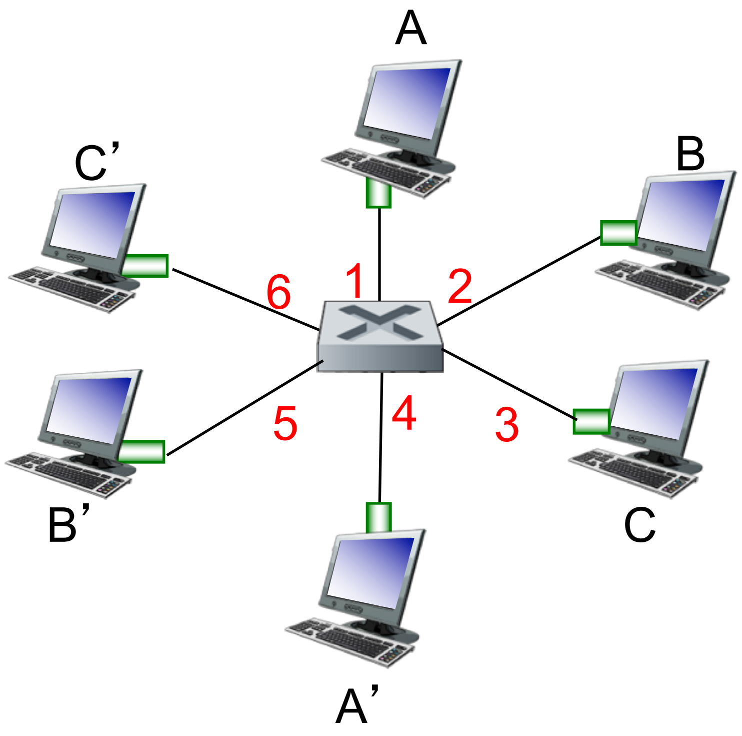


Ethernet switch

- link-layer device: takes an *active* role in forwarding frames
  - store, forward Ethernet frames
    - examine incoming frame's MAC address,
    - selectively forward frame to one-or-more outgoing links
    - when frame is to be forwarded on segment, uses CSMA/CD to access segment
- *transparent*
  - hosts are unaware of presence of switches
- *plug-and-play, self-learning*
  - switches do not need to be configured, i.e., they create and manage their own forwarding tables

Switch: *multiple* simultaneous transmissions

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, but no collisions; full duplex
  - each link is its own collision domain
- *switching:* A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six interfaces (1,2,3,4,5,6)

Switch forwarding table

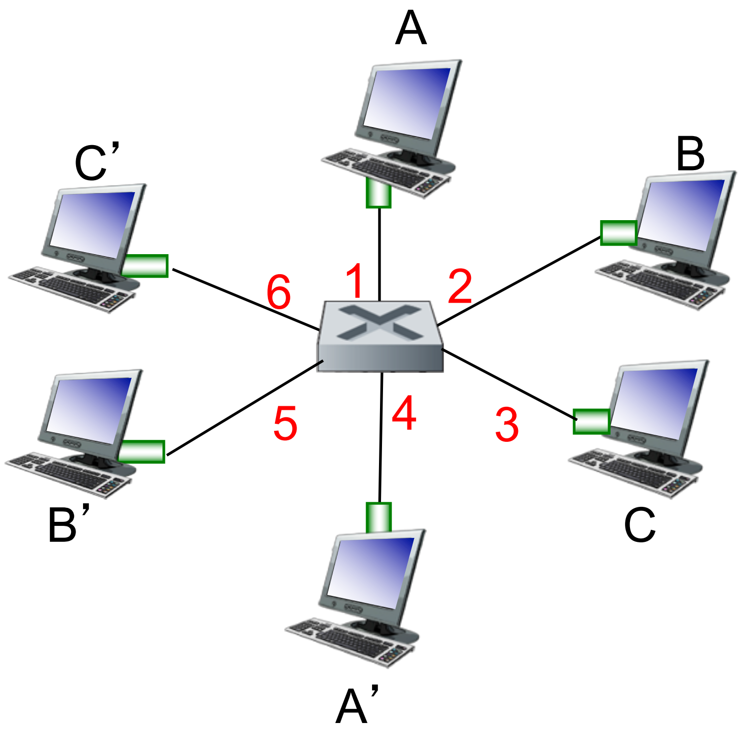
Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

A: each switch has a forwarding (switch) table, each entry consist of:

- MAC address of host, interface to reach host, time stamp
- looks like a routing table!

Q: how are entries created, maintained in forwarding table?

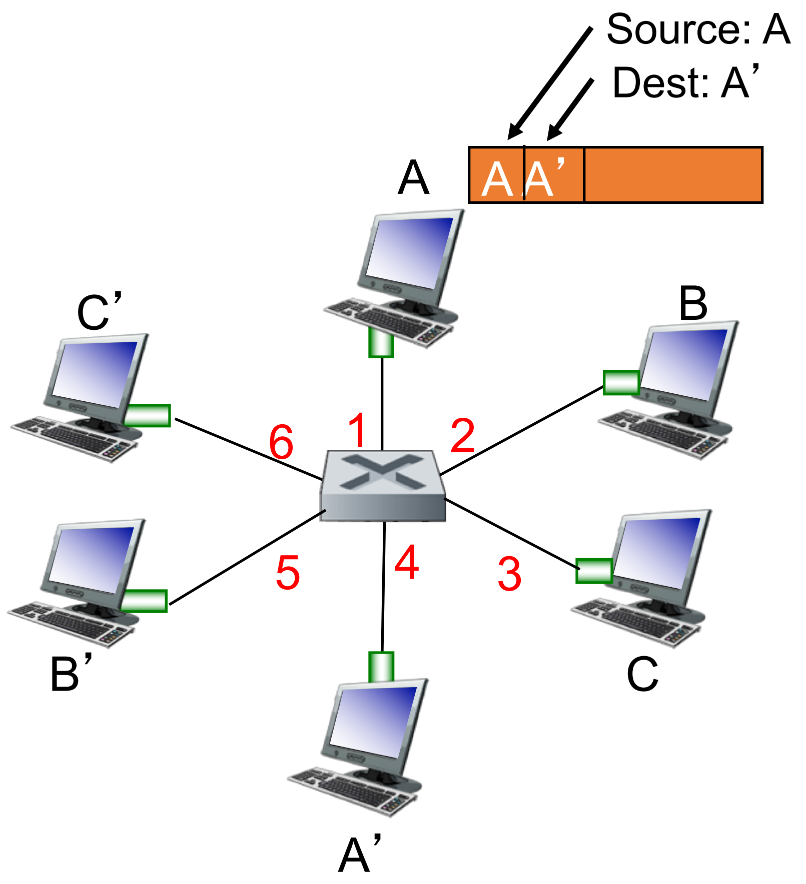
- uses a backwards learning algorithm (something like a routing protocol) to populate table with forwarding entries



switch with six interfaces (1,2,3,4,5,6)

Switch: self-learning

- switch *learns* which hosts can be reached through which interfaces
  - when frame received, switch "learns" location of sender: incoming LAN segment
  - records sender/location pair in switch table

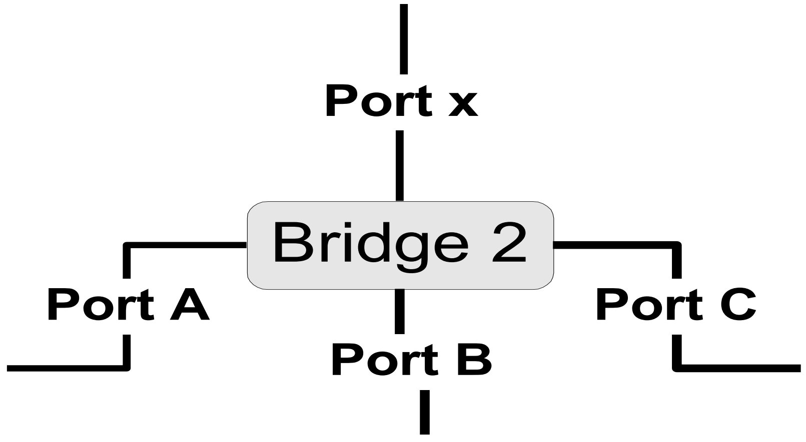


MAC addr	interface	TTL
A	*1*	*60*

Switch table (initially empty)

Frame Forwarding

- assume a MAC frame arrives on port X



Is MAC address of destination in forwarding database for ports A, B, or C ?

Found?

Forward the frame on the stored port

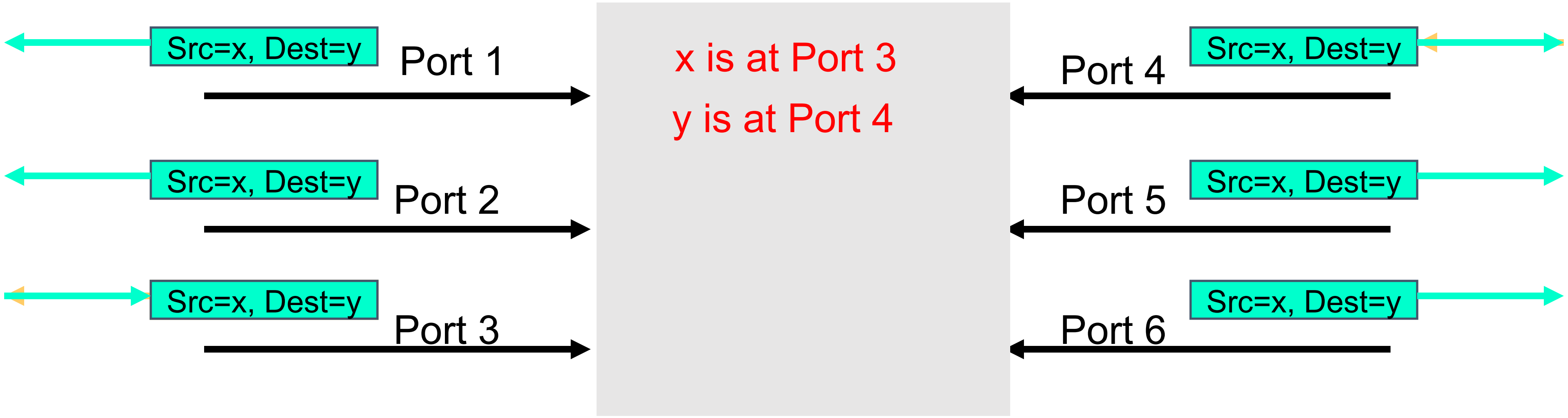
Not found ?

Flood the frame

send the frame on all ports except port X

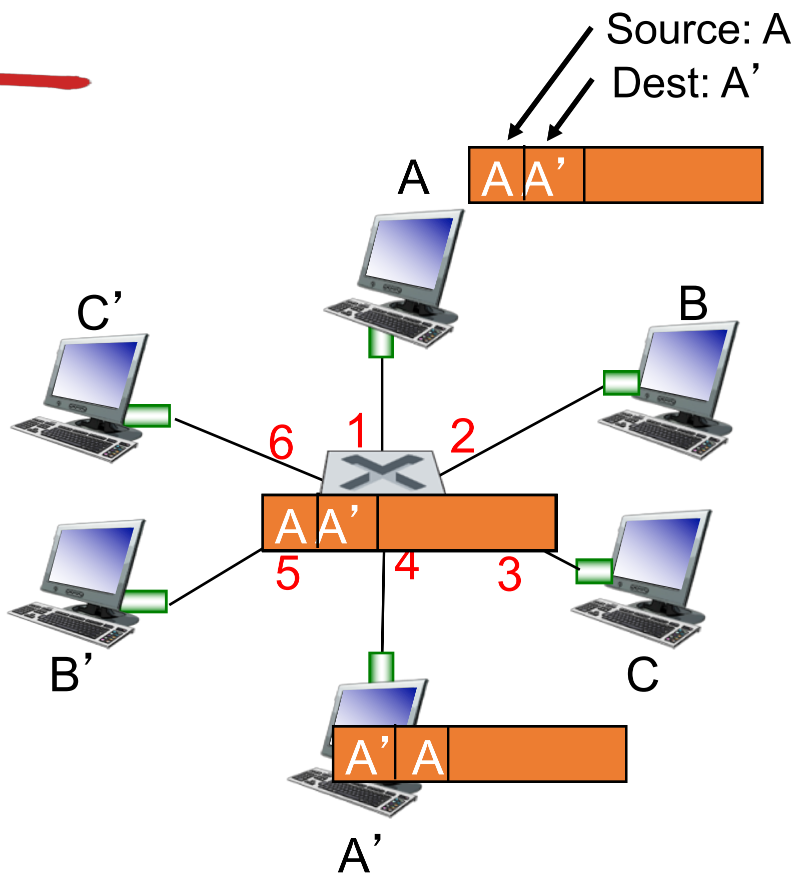
Self Learning (Backwards Learning)

- routing tables entries are set automatically with a simple heuristic:
  - the source address field of a frame that arrives on a port informs switch that this address (host) is reachable from this port



Self-learning, forwarding: example

- frame destination, A’, location unknown: *flood*
  - destination A location known: selectively send on just one link



MAC addr	interface	TTL
A	*1*	*60*
A’	*4*	*60*

switch table (initially empty)

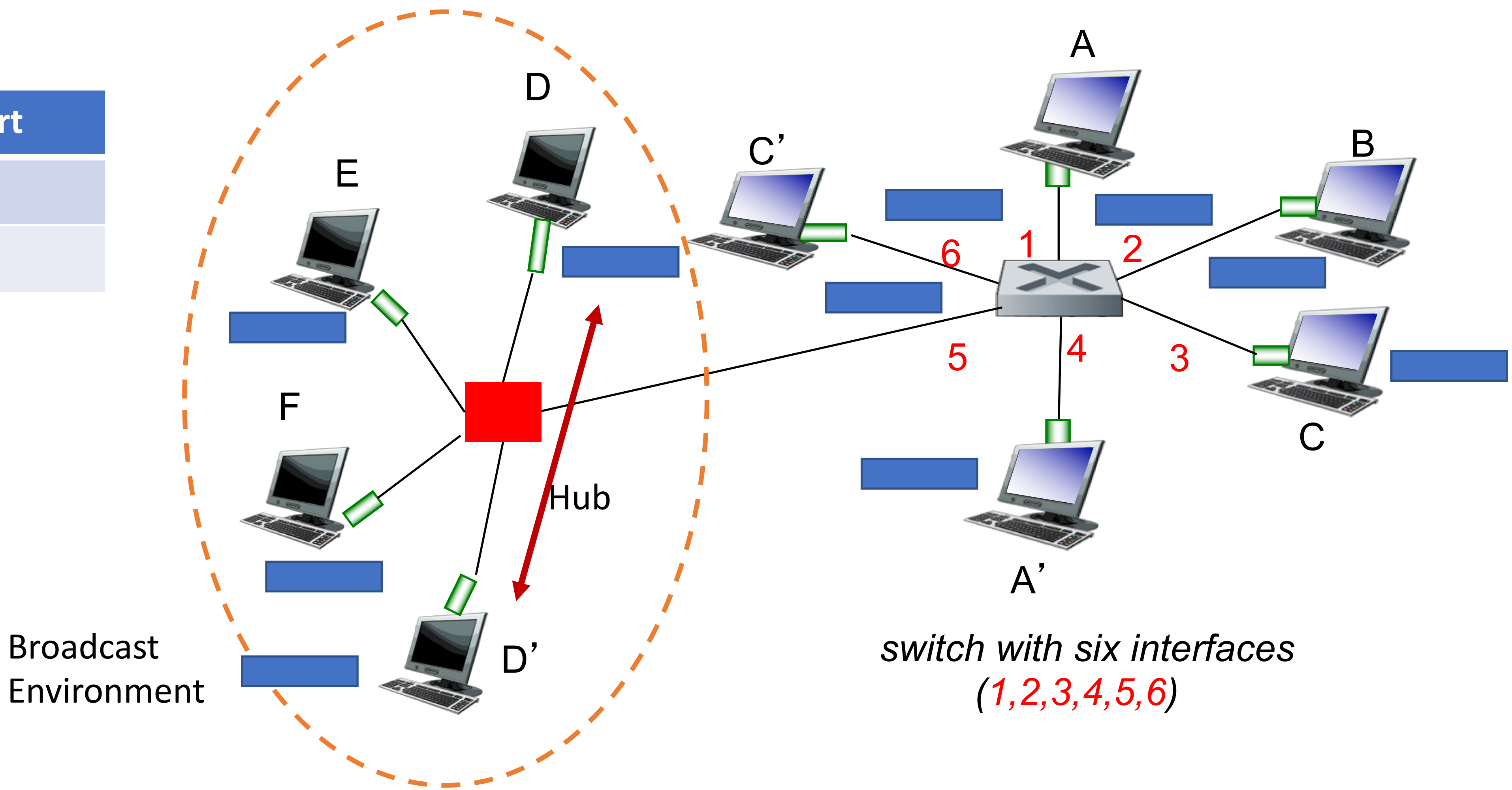
Learning Bridges Algorithm

For each frame received:

- the bridge enters the source mac address and port in forwarding table or refreshes timer of an existing entry (address seen previously and not yet expired)
- the bridge looks to find entry for destination MAC address in forwarding database
  - if port/interface on which frame is received is same as that for destination, drop frame (e.g., broadcast environment, see next slide)
  - if an entry for destination MAC address exists in forwarding table, reset timer and forward the frame
  - if entry not found, the bridge floods all ports with the frame except for port on which the frame was received
- all entries are deleted after some time (default is 15 seconds).

Switch & Broadcast Environment

Address	Port
D	5
D'	5



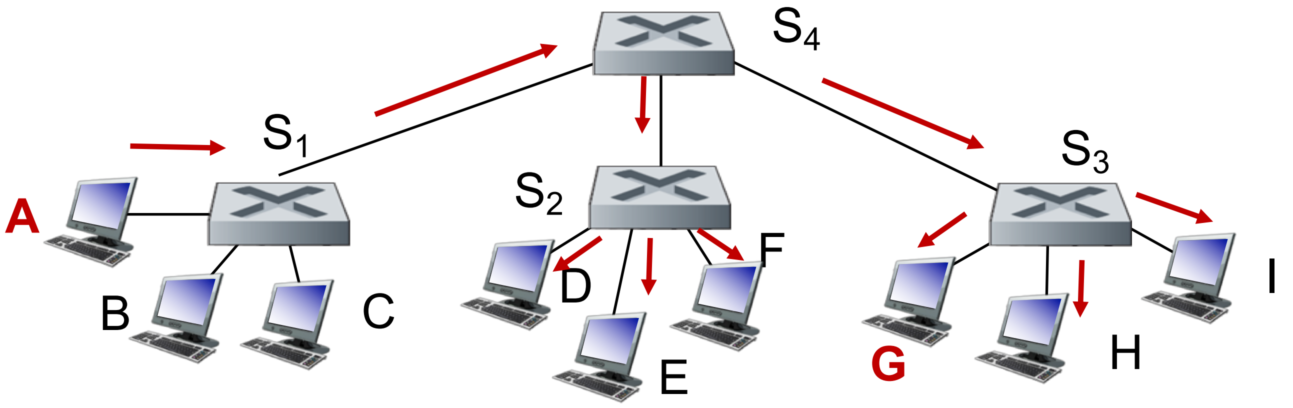
Switch: frame filtering/forwarding

when frame received at switch:

1. record incoming link, MAC address of sending host
2. index switch table using MAC destination address
3. if entry found for destination
   then {
   if destination on segment from which frame arrived (i.e., same segment)
      then drop frame
      (i.e, destination and source in same direction, no need to forward, see D-D' next slide)
      else forward frame on interface indicated by entry
   }
   else flood /* forward on all interfaces except arriving
                   interface */

Interconnecting switches

self-learning switches can be connected together:



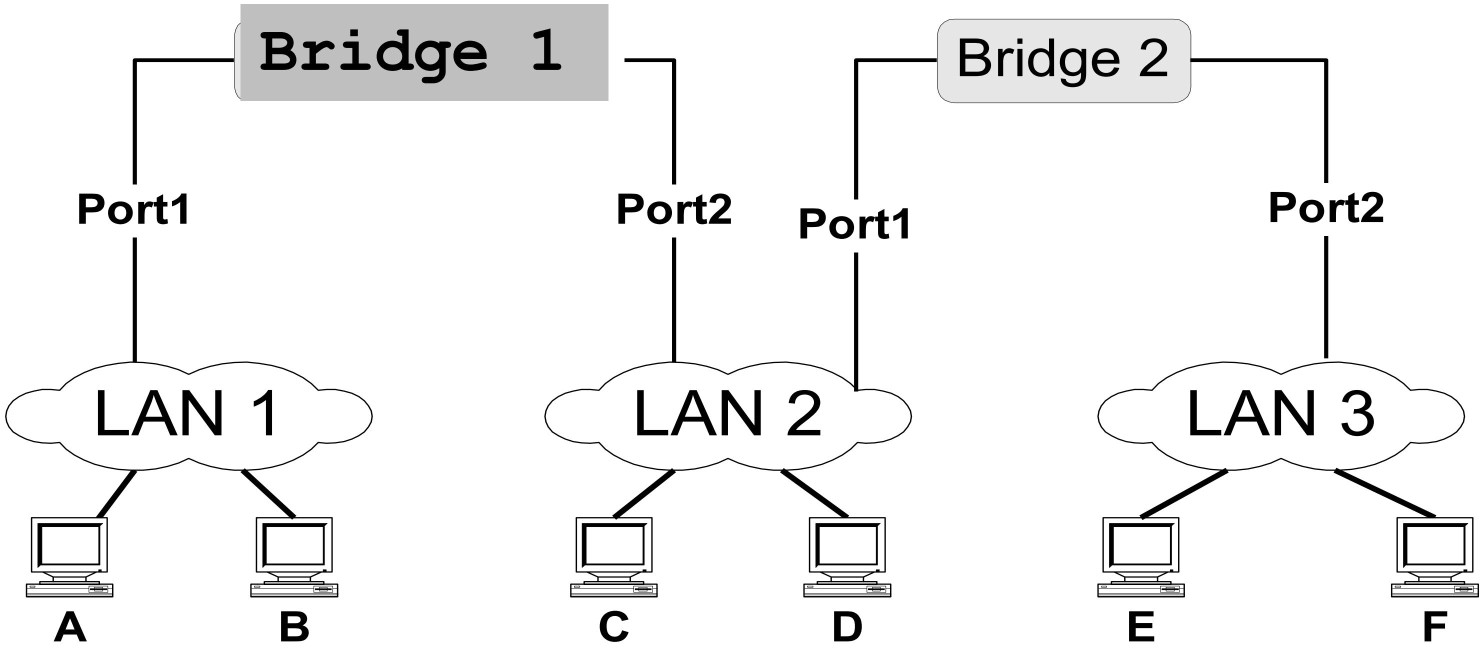
Q: sending from A to G - how does S_1 know to forward frame destined to G via S_4 and S_3?

A: self learning! (works exactly the same as in single-switch case!)

Example

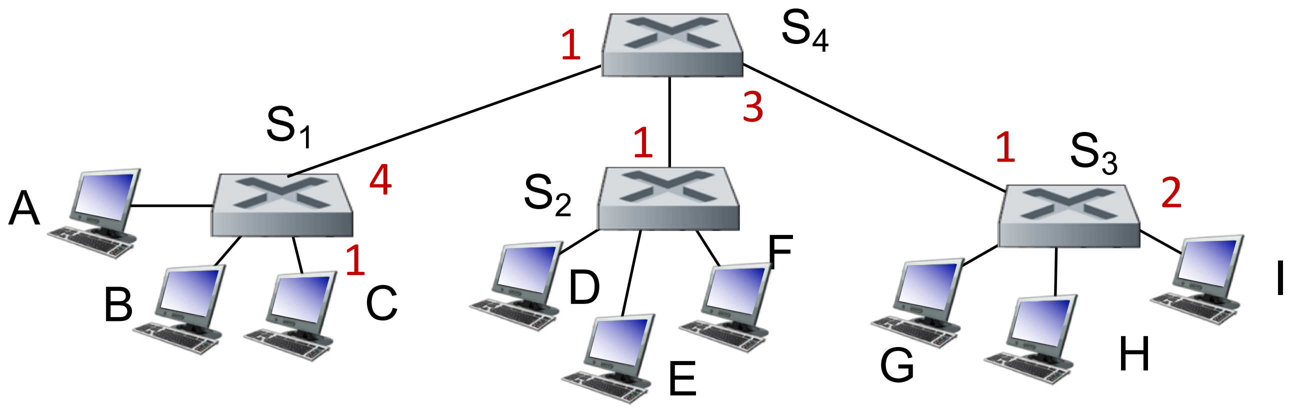
- the following packets have been transmitted in the two bridge network below:
  (Src=A, Dest=F), (Src=C, Dest=A), (Src=E, Dest=C)

- After the 3 transmissions: what have the bridges learned?



Self-learning multi-switch example

Suppose C sends frame to I, I responds to C



S1

Address	Port
C	1
I	4

S4

Address	Port
C	1
I	3

S3

Address	Port
C	1
I	2

- Q: show switch tables and packet forwarding in S_1, S_2, S_3, S_4

S2

Address	Port
C	1

Loops and Routing